Supplementary Material

How Much of Genomic Language Model Variant-Effect Prediction Is Base Composition?

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This supplement provides the complete result tables and implementation details underlying the main paper's figures and Table 1. Code and released artifacts are available at `github.com/rikhinkavuru/composit`

Implementation. Composition features (C1/C2) are computed from hg38 with the Carlson pentanucleotide neutral-rate table; the 19 features are the Δ_k ($k \in \{1, 2, 3\}$) and ΔGC values at windows $w \in \{11, 21, 51\}$, the pentanucleotide rate, reference and alternate CpG indicators, a transition indicator, and flanking GC. The composition classifier is a histogram gradient-boosted-tree model (250 trees, learning rate 0.05) trained leave-one-chromosome-out. The one-hot CNN uses ± 100 bp windows (reference with the alternate substituted at the centre), two Conv1d layers (32 channels, width 8), global max-pool, and a linear head, trained with Adam under chromosome-grouped folds. The pentanucleotide calibration subtracts a per-context neutral score estimated leave-one-chromosome-out from control variants, or, in the label-free variant, from a genome-wide table of 3,072 contexts and 3.6×10^7 sampled GPN-MSA scores. All model scores are authors' released per-variant values; no gLM inference is run locally.

Table 1: Complete leave-one-chromosome-out AUROC, AUPRC, composition recovery ratio, and Benjamini–Hochberg-adjusted DeLong p (model vs composition) for all methods on every benchmark. Recovery ratios above one (>1) mean the method does not exceed the composition baseline.

Method	AUROC	AUPRC	Recovery	DeLong p
ClinVar (composition-only AUROC 0.720)				
CADD	0.986	0.958	0.45	$< 10^{-16}$
Evo2-40B	0.963	0.870	0.47	$< 10^{-16}$
GPN-MSA(csv)	0.961	0.937	0.48	$< 10^{-16}$
Evo2-7B	0.960	0.882	0.48	$< 10^{-16}$
GPN-MSA	0.958	0.861	0.48	$< 10^{-16}$
phyloP-447w	0.912	0.757	0.53	$< 10^{-16}$
phyloP-100w	0.912	0.775	0.53	$< 10^{-16}$
SpliceAI	0.667	0.502	1.32	5e-41
TraitGym-mendelian (composition-only AUROC 0.586)				
GPN-MSA	0.904	0.694	0.21	2e-68
CADD	0.893	0.712	0.22	2e-50
phastCons	0.858	0.527	0.24	7e-40
phyloP-100v	0.857	0.655	0.24	2e-34
Evo2-40B	0.851	0.557	0.25	5e-45
GPN	0.718	0.422	0.40	1e-08
Evo2-7B	0.710	0.385	0.41	6e-08
SpeciesLM	0.620	0.201	0.72	1e-01
NT-2.5B	0.534	0.120	>1	3e-02
HyenaDNA	0.513	0.115	>1	3e-03
Caduceus	0.509	0.108	>1	1e-03
TraitGym-complex (composition-only AUROC 0.520)				
phastCons	0.617	0.237	0.17	4e-13
CADD	0.616	0.250	0.18	8e-13
phyloP-100v	0.583	0.184	0.24	2e-06
GPN-MSA	0.580	0.212	0.25	1e-05
Caduceus	0.518	0.114	1.13	8e-01
Evo2-40B	0.516	0.137	1.26	7e-01
NT-2.5B	0.516	0.114	1.27	7e-01
GPN	0.510	0.112	>1	4e-01
HyenaDNA	0.508	0.111	>1	3e-01
Evo2-7B	0.505	0.118	>1	3e-01
SpeciesLM	0.504	0.110	>1	2e-01
GLRB-eQTL (composition-only AUROC 0.596)				
GPN-MSA	0.560	0.599	>1	4e-03

Table 2: E1 partial Spearman correlation between |LLR| and each context-free confounder (controlling for consequence): 5-mer mutation rate μ_5, reference CpG, ΔGC, 3-mer spectrum shift Δ_3, and flanking GC.

Method	μ_5	CpG	ΔGC	Δ_3	flankGC
ClinVar					
CADD	-0.06	-0.04	-0.03	+0.05	-0.04
Evo2-40B	-0.18	-0.12	-0.11	+0.03	-0.04
Evo2-7B	-0.13	-0.09	-0.07	+0.05	-0.06
GPN-MSA	-0.33	-0.31	-0.14	-0.00	-0.11
GPN-MSA(csv)	-0.27	-0.24	-0.13	+0.02	-0.12
SpliceAI	-0.04	-0.05	-0.02	+0.03	-0.06
phyloP-100w	+0.04	+0.06	+0.02	+0.05	-0.00
phyloP-447w	+0.16	+0.19	+0.05	+0.07	+0.05
TraitGym-mendelian					
CADD	-0.02	-0.00	-0.01	-0.01	+0.23
Caduceus	+0.07	+0.08	+0.10	+0.11	+0.06
Evo2-40B	-0.11	+0.00	-0.07	+0.01	+0.20
Evo2-7B	-0.03	+0.01	-0.01	+0.06	+0.07
GPN	+0.06	+0.13	+0.04	+0.12	+0.18
GPN-MSA	-0.16	-0.09	-0.10	-0.00	+0.04
HyenaDNA	-0.01	-0.03	+0.04	+0.01	-0.01
NT-2.5B	+0.11	+0.12	+0.10	+0.13	+0.04
SpeciesLM	+0.03	+0.08	+0.06	+0.10	+0.10
phastCons	-0.07	-0.08	-0.03	-0.02	-0.00
phyloP-100v	+0.01	+0.04	-0.04	+0.01	+0.05
TraitGym-complex					
CADD	-0.02	-0.03	+0.01	-0.01	+0.02
Caduceus	+0.01	+0.11	+0.00	+0.10	+0.01
Evo2-40B	-0.08	+0.03	-0.06	+0.06	+0.05
Evo2-7B	-0.05	+0.05	-0.05	+0.08	-0.00
GPN	+0.15	+0.26	+0.07	+0.13	+0.14
GPN-MSA	-0.20	-0.15	-0.09	-0.01	-0.12
HyenaDNA	-0.04	-0.01	-0.04	+0.02	+0.01
NT-2.5B	+0.04	+0.14	+0.01	+0.12	-0.01
SpeciesLM	+0.03	+0.14	+0.01	+0.11	+0.03
phastCons	-0.14	-0.19	-0.04	-0.02	-0.11
phyloP-100v	+0.06	+0.09	+0.02	+0.02	+0.10

Table 3: E4 within-element Kircher saturation MPRA: raw versus partial Spearman of GPN-MSA LLR against measured $\log_2$ effect (partial controls for substitution type, ΔGC, and CpG), per element. The drop (raw−partial) is near zero throughout.

Element	n	raw ρ	partial ρ	drop
SORT1	5379	-0.127	-0.161	+0.035
TERT	3108	+0.185	+0.165	+0.021
ZRS	2910	-0.018	-0.040	+0.022
PKLR	2814	+0.097	+0.096	+0.000
LDLR	1908	+0.397	+0.403	-0.005
FOXE1	1800	-0.033	-0.023	-0.010
BCL11A	1799	+0.033	+0.030	+0.004
RET	1797	+0.131	+0.129	+0.002
MYCrs6983267	1792	+0.017	+0.024	-0.007
IRF6	1785	+0.193	+0.207	-0.013
MSMB	1773	+0.105	+0.089	+0.016
TCF7L2	1770	-0.017	-0.025	+0.008
UC88	1761	-0.010	-0.005	-0.005
ZFAND3	1736	+0.199	+0.199	-0.000
MYCrs11986220	1389	+0.041	+0.041	-0.000
IRF4	1353	+0.216	+0.217	-0.002
GP1BA	1154	+0.149	+0.121	+0.028
F9	904	+0.056	+0.063	-0.007
HNF4A	855	+0.057	+0.066	-0.009
HBG1	822	+0.235	+0.254	-0.019
HBB	561	+0.332	+0.336	-0.004

Table 4: One-hot CNN baseline (trained from scratch on ±100 bp windows): it stays below both the composition baseline and the gLMs on every benchmark.

Benchmark	AUROC	AUPRC
TraitGym-mendelian	0.516	0.099
TraitGym-complex	0.537	0.113
ClinVar	0.561	0.197